A resonance-scale spectral law for a deterministically kicked Pell cat map

WoW research programme

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Status and main result

This is a project derivation with a complete argument below, not a claim of priority, external referee acceptance, or full Lean formalization. The result concerns the actual kicked quantum propagator, but its **K-th power at the arithmetic recurrence scale**, not microscopic nearest-neighbour statistics. The critical quantum kick tends to zero. A fixed nonzero classical perturbation is not covered.

Put

$$\alpha = 1 + \sqrt{2}, \quad \lambda = \alpha^2, \quad T = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}.$$

Let K tend to infinity through odd primes, and define positive integers N,F by

$$N + F\sqrt{2} = \alpha^K, \qquad N^2 - 2F^2 = -1.$$

The moduli N may be composite. On the odd subspace of the standard N-dimensional torus quantization, let U_N be the phase-normalized quantization of T described in Section 1, and let

$$B_N f(x) = 2\cos(2\pi x/N) f(x), \quad U_{\varepsilon,N} = e^{i\varepsilon B_N} U_N, \quad d_N = (N-1)/2.$$

Theorem A (critical resonance law). For every fixed real tau and every integer m, with epsilon=tau/sqrt(2K),

$$\lim_{K \longrightarrow \infty} \frac{1}{d_N} \operatorname{Tr}_- U_{\varepsilon,N}^{mK} = \exp(-m^2\tau^2/2). \tag{A}$$

Consequently the empirical eigenvalue measure of U_(epsilon,N)^K converges weakly on the unit circle to the wrapped normal law. For tau nonzero its angular density on [-pi,pi) is

$$w_\tau(\theta) = \frac{1}{\sqrt{2\pi}|\tau|} \sum_{\ell \in \mathbb{Z}} \exp\left[-\frac{(\theta + 2\pi\ell)^2}{2\tau^2} \right]. \tag{B}$$

For tau=0 the limit is the point mass at 1. In particular the normalized resonance echo obeys

$$d_N^{-2} \left| \operatorname{Tr}_- U_{\varepsilon,N}^{mK} \right|^2 \longrightarrow e^{-m^2\tau^2}. \tag{C}$$

This is a deterministic limit: there is no parameter average in (A).

The principal finite estimate, proved before any limit is taken, is

$$\left\| V_\varepsilon U_{\varepsilon,N} V_\varepsilon^* - e^{i\varepsilon\overline{B}} U_N \right\|_{2,-} \le \frac{12\varepsilon^2}{\sqrt{K}}, \qquad K \ge 11, \tag{D}$$

for an explicit unitary V_epsilon. Here the norm is normalized Hilbert–Schmidt norm on the entire odd subspace. It does not control an individual level or an operator-norm spectral gap. The finite comparison error in (A) is at most 6|m|tau^2/sqrt(K); an additional Gaussian characteristic-function convergence error is o(1), not assigned that rate.

1. Arithmetic and the fixed classical observable

The recurrence (N_j,F_j)=(N_(j-1)+2F_(j-1),N_(j-1)+F_(j-1)), starting at (1,0), shows that N is odd. Put, in Z/NZ,

$$s = F^{-1}, \qquad s^2 = 2, \qquad h = 3 + 2s, \qquad q = -F.$$

Since (1+s)^K=N+Fs=1, h^K=1. Also h-1=2(1+s) is a unit. For every prime ell dividing N, the order of h modulo ell is therefore precisely K: K is prime and that order is not 1. Hence 1-h^j is a unit for 1<=j<K, and 1+h^j is a unit for every j, because K is odd. In particular ell=1 mod K, and N=1 mod 2K. Neither squarefreeness nor primality of N is required.

Use the counting-measure inner product and centered Weyl operators

$$W(a,b)f(x) = e_N(a(x - b/2))f(x - b), \qquad S_u f(x) = f(u^{-1}x), \qquad Rf(x) = f(-x).$$

Divisions in Weyl exponents mean inverses in Z/NZ. The multiplication and covariance formulas are

$$W(a,b)W(c,d) = e_N((ad - bc)/2)W(a + c, b + d),$$
$$S_u W(a,b) S_u^* = W(u^{-1}a, ub), \qquad RW(v)R = W(-v).$$

Individual Weyl translations need not preserve parity; their symmetrized sums do. On the odd subspace take

$$U_0 = S_h^{-1}, \qquad B = W(q,1) + W(q,1)^*, \qquad U_\varepsilon = e^{i\varepsilon B} U_0. \tag{1}$$

In these coordinates U_0^K=I exactly.

For completeness this is unitarily equivalent to the fixed-map formulation in Theorem A, including the kick, not merely to its compressed observable. Let C_t f(x)=e_N(tx^2/2)f(x) and let F_N have kernel e_N(xy)/sqrt(N). Set

$$M = \begin{pmatrix} -1/(2s) & 1 \\ -1/2 & -s \end{pmatrix}, \qquad V = C_{s^{-1}} S_{1/2} F_N C_{2s}.$$

Its factors have Weyl covariance matrices respectively

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} u^{-1} & 0 \\ 0 & u \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Multiplication gives covariance M, det M=1, and T=M diag(h,h^(-1)) M^(-1). Define U_N=VU_0V. *The index of V*W(1,0)V is M^(-1)(1,0)=(-s,1/2). Further conjugation by S_2, which commutes with U_0, changes this to (-s/2,1)=(-F,1) modulo N. Thus S_2 V* conjugates the entire physical kicked propagator to (1). These unitaries all preserve parity.

With the usual semiclassical convention e^(i action/hbar), the classical kick amplitude corresponding to epsilon is proportional to hbar*epsilon. Here hbar is proportional to 1/N. At the critical scale this amplitude is of order 1/(N sqrt(K)), not a fixed classical perturbation.

2. Two summable phase estimates

Write C(A)=U_0 A U_0*, and

$$B_j = C^j(B) = W(v_j) + W(-v_j), \qquad v_j = (qh^j, h^{-j}), \qquad \overline{B} = K^{-1} \sum_{j=0}^{K-1} B_j. \tag{2}$$

The conjugation C shifts the label cyclically, so C(bar B)=bar B. Define

$$a_j = \left| e_N(q(h^j - h^{-j})) - 1 \right|, \qquad c_j = \frac{K - 1 - 2j}{2K}, \qquad G = \sum_{j=0}^{K-1} c_j B_j. \tag{3}$$

Then a_0=0, a_(K-j)=a_j. Direct subtraction of the coefficients gives the exact cohomological equation

$$G - C(G) = B - \overline{B}. \tag{4}$$

The coefficients have zero mean. Importantly, they are small at the half-period at which the large commutator phases accumulate.

Let P_t denote the second Pell sequence, alphaˆt=Q_t+P_t sqrt(2). For the signed representative delta of j mod K one has

$$q(h^j - h^{-j}) = -2P_{2\delta} \pmod N. \tag{5}$$

Indeed $h^{\text{delta=(3+2s)}}$delta=Q_(2delta)+P_(2delta)s, and F*s=1. For delta=(K-1)/2-ell positive,

$$\frac{P_{2\delta}}{N} = \frac{\lambda^{-\ell}}{\sqrt{2}\,\alpha}\frac{1-\lambda^{-2\delta}}{1-\lambda^{-K}}. \tag{6}$$

Put r=lambdaˆ(-1), A=4pi/(sqrt(2)alpha). Using |exp(it)-1|<=|t| and summing both signed representatives proves

$$L_K := \sum_{j=0}^{K-1} a_j \le \frac{2A}{(1-r)(1-\lambda^{-K})} = \frac{2\sqrt{2}\pi}{1-\lambda^{-K}} < 9 \quad (K \ge 3). \tag{7}$$

There is a sharper weighted estimate. At j=(K-1)/2-ell the weights at j and K-j have absolute values ell/K and (ell+1)/K. Therefore

$$\sum_j |c_j| a_j \le \frac{A}{K(1-\lambda^{-K})} \sum_{\ell \ge 0} (2\ell+1) r^\ell = \frac{2\pi}{K(1-\lambda^{-K})}. \tag{8}$$

The final identity follows from A(1+r)/(1-r)ˆ2=2pi. The estimates are analytic inequalities; the regression's decimal phase values are only finite checks of them.

3. Quadratic lifting and orthogonal commutator supports

Lemma 1 (lifting). Fix m. For all sufficiently large K, every signed m-term reciprocal relation

$$\sum_{\nu=1}^{m} \epsilon_\nu h^{j_\nu} = \sum_{\nu=1}^{m} \epsilon_\nu h^{-j_\nu} = 0 \pmod N, \qquad \epsilon_\nu \in \{-1, 1\}, \tag{9}$$

can be cyclically rotated to a polynomial P with P(lambda)=0. Its coefficient l1 norm is at most m and its degree is at most K-ceil(K/m). For m=4 this holds for every odd prime K>=11, and the relation is then coefficientwise balanced.

Proof. Cancel opposite signs at identical labels. If anything remains, cut the cyclic support at a largest gap and rotate. There are at most m support points, giving the stated degree bound. Both congruences in (9) survive. Divide P by the monic polynomial Zˆ2-6Z+1 over the integers. Its remainder aZ+b vanishes at h and hˆ(-1); their difference 4s is a unit. Hence N divides a and b. A nonzero quadratic norm of P(lambda) is thus an integer divisible by Nˆ2, but

$$|P(\lambda)P(\lambda^{-1})| \le m^2 \lambda^{K - \lceil K/m \rceil} < N^2 \tag{10}$$

for sufficiently large K at fixed m. This is impossible, so P(lambda)=0. For m=4 a sufficient condition is $\text{lambda}^{(K/4)(1-\text{lambda}}$(-K))ˆ2>64. It holds for K>=11: lambda>5, 5ˆ(11/4)>80, and 1-5ˆ(-11)>9/10 already give a strict lower bound 64.8. A nonzero polynomial of l1 norm at most 4 cannot vanish at lambda>3: its leading term strictly dominates the total of all lower terms. Thus in this case the polynomial was zero. End of proof.

It follows that the 2K(K-1) vectors

$$\epsilon v_t + \delta v_s, \qquad 0 \le t < s < K, \quad \epsilon, \delta \in \{-1, 1\}, \tag{11}$$

are nonzero and distinct for K>=11. Any collision would give a balanced four-letter relation, forcing precisely the original unordered signed pair. The matrices W(v) form an orthonormal basis for the full normalized Hilbert–Schmidt inner product, also for composite odd N, because Tr W(v)=N times the indicator v=0.

Each of the four Weyl coefficients in [B_t,B_s] has modulus a_(s-t). Different unordered time pairs have disjoint support by (11). Since c_t-c_s=(s-t)/K, exact orthogonality gives

$$\|[G, \overline{B}]\|_{2,N}^2 = \frac{4}{K^4} \sum_{j=1}^{K-1} (K-j) j^2 a_j^2 = \frac{2}{K^3} \sum_{j=1}^{K-1} j(K-j) a_j^2 \le \frac{9}{K}. \tag{12}$$

Here ||A||_(2,N)^2=Tr(A*A)/N. The middle identity uses a_(K-j)=a_j; the last uses j(K-j)<=K^2/4, a_j^2<=2a_j, and (7).

The operator norm needs no orthogonality. Four Weyl coefficients, (7), and (8) give

$$\|[G, B]\|_{\mathrm{op}} < \frac{26}{K}, \qquad \|[B, \overline{B}]\|_{\mathrm{op}} < \frac{36}{K}. \tag{13}$$

For a parity-preserving A, positivity of A*A implies

$$\|A\|_{2,-} \le \sqrt{\frac{2N}{N-1}}\, \|A\|_{2,N}. \tag{14}$$

No orthogonality of Weyl operators restricted to the odd subspace is assumed.

4. A finite normal form without a small generator

For finite Hermitian A,D, real epsilon, and any unitarily invariant norm,

$$\|e^{i\varepsilon A} e^{i\varepsilon D} - e^{i\varepsilon (A+D)}\| \le \frac{\varepsilon^2}{2} \|[A, D]\|. \tag{15}$$

This is the usual two-factor commutator estimate, not a new general product formula. To prove it, differentiate exp(i(epsilon-t)(A+D)) exp(itA) exp(itD). After factoring unitaries, the integrand is exp(itA)D exp(-itA)-D. Integrate its derivative once more; the resulting triangular integration domain has area epsilon^2/2. Negative epsilon follows by replacing A,D by -A,-D. In particular there is no assumption that |epsilon|*||A|| is small.

Set V_epsilon=exp(-i epsilon G). By (4),

$$V_\varepsilon U_\varepsilon V_\varepsilon^* = e^{-i\varepsilon G} e^{i\varepsilon B} e^{i\varepsilon C(G)} U_0.$$

Apply (15) first to -G,B, then to B-G,C(G). Their total sum is bar B, and [B-G,C(G)]=[B-G,bar B]. Equations (12)–(14) show

$$\begin{aligned} \|V_\varepsilon U_\varepsilon V_\varepsilon^* - e^{i\varepsilon \overline{B}} U_0\|_{2,-} &\le \frac{\varepsilon^2}{2} \left(\frac{62}{K} + 3\sqrt{\frac{2N}{(N-1)K}} \right) \\ &\le \frac{12\varepsilon^2}{\sqrt{K}}, \qquad K \ge 11. \end{aligned} \tag{16}$$

For the last step it even suffices to use N>=7 and 31/sqrt(11)+(3/2)sqrt(7/3)<12. This proves (D) for all real epsilon. G may have operator norm of order K. Expanding its exponential in

powers of epsilon and requiring epsilon*K small would incorrectly discard the critical regime. Estimate (15) avoids that requirement.

Since bar B commutes with U_0 and U_0^K=I, telescoping unitary powers in the same norm yields, for any integer m,

$$\left|\tau_-(U_\varepsilon^{mK}) - \tau_-(e^{im\varepsilon K\overline{B}})\right| \le 12|m|\varepsilon^2\sqrt{K}, \qquad \tau_-(A) = d_N^{-1}\,\mathrm{Tr}_-\,A. \tag{17}$$

For negative m use adjoints. Cauchy–Schwarz for the normalized trace justifies the passage from the norm to traces.

5. A self-contained global law for the averaged observable

The remaining input is established here directly on the whole odd subspace; no unproved claim about the old arithmetic-moment corpus is imported. Define

$$X_K = \sqrt{K/2}\,\overline{B} = (2K)^{-1/2}\sum_{j=0}^{K-1} B_j.$$

Lemma 2 (Gaussian average). For each fixed integer r>=1,

$$\tau_-(X_K^{2r}) = (2r-1)!! + O_r(K^{-1}) + O_r(K^r/N), \tag{18}$$

and for each fixed odd m, eventually

$$|\tau_-(X_K^m)| \le (2K)^{m/2}/(N-1). \tag{19}$$

The empirical measures of X_K converge weakly and in quadratic Wasserstein distance to the standard normal law.

Proof. In the full Weyl representation Tr(RW(v))=1 for every v: the unique fixed point x=-b/2 of RW(a,b) contributes phase 1. Thus the normalized odd trace of a word e_N(phi)W(v) is

$$e_N(\phi)\frac{N\mathbf{1}_{v=0}-1}{N-1}. \tag{20}$$

Expanding X_K^m, the noncentral part is bounded by (2K)^(m/2)/(N-1). The central part is the weighted count of the reciprocal relations (9), with normalization N/(N-1) times (2K)^(-m/2).

Here is the required count including structurally forced nonbalanced relations. For a polynomial of l1 norm at most m vanishing at lambda, a gap d>2 log_lambda m separates two independently vanishing pieces. Indeed write P=P_low+Z^(a+d)P_high with deg P_low<=a and nonnegative exponents in P_high. If P_high(lambda) is nonzero, its algebraic-integer norm is a nonzero integer. Since |P_high(lambda^(-1))|<=m, |P_high(lambda)|>=1/m. Cancellation with |P_low(lambda)|<=m lambda^a would force lambda$^{d<=m}$2, a contradiction.

Repeatedly splitting gives components with span bounded in terms of m, for example by D_m=(m-1)(ceil(2 log_lambda m)+1). Every nonzero component has at least three signed occurrences: two distinct positive powers cannot cancel with signs of magnitude one. Restore previously cancelled opposite pairs as two-occurrence components. A nonbalanced 2r-letter word therefore has at most r-1 components. For each of finitely many partitions of the 2r positions, signs and bounded relative offsets, each component has at most K translations on the cyclic time set. There are only O_r(K^(r-1)) such words. There is no additional factor K from choosing a cut: the count is over the component translations themselves, and the largest-gap cut was used only to prove bounded width. Their normalized contribution is O_r(1/K).

For odd m, a lifted P is an integer multiple of Z^2-6Z+1, so P(1) is divisible by -4. But P(1) is the sum of an odd number of signs. Thus no central word exists once Lemma 1 applies, proving (19).

A balanced 2r-word pairs identical time labels with opposite signs. Assignments with repeated pair labels cause only O_r(K^(r-1)) overcount. For every matching of the 2r positions there are K^r choices of pair labels and 2^r orientations. The Weyl phase of such a word is a product, over the crossing pairs, of factors e_N(q(h^{a-h}(-a))) or their conjugates. This follows by summing the symplectic products of the signed indices in word order: disjoint or nested pairs cancel; a crossing pair retains one full symplectic product.

Each crossing difference of independent uniform pair labels is uniform mod K. Independence of the different crossing factors is neither true in general nor needed. The elementary inequality |product u_l-1|<=sum |u_l-1| for |u_l|=1 and (7) give an average phase error at most 9 times the number of crossings divided by K. Hence each matching contributes 1+O_r(1/K) after normalization. There are (2r-1)!! matchings. Together with the nonbalanced count and (20), this proves (18).

The second-moment bound gives tightness. Higher even moments give uniform integrability when passing moments to subsequential limits. The Gaussian moment sequence is determinate, for example by Carleman's criterion, since ((2r-1)!!)^(1/(2r))<=sqrt(2r). Every subsequential limit is therefore standard normal. Convergence of the second moments upgrades weak convergence to quadratic Wasserstein convergence. End of proof.

6. Critical and subcritical propagator spectra

For epsilon=tau/sqrt(2K), (17) becomes

$$\left|\tau_{-}(U_{\varepsilon}^{mK}) - \tau_{-}(e^{im\tau X_K})\right| \le \frac{6|m|\tau^2}{\sqrt{K}}. \tag{21}$$

Lemma 2 and bounded continuity of exp(im tau x) give (A). Integer Fourier coefficients determine measures on the circle: trigonometric polynomials are dense in continuous functions there. Applying the characteristic function of a normal variable proves (B), and taking absolute squares proves (C). No eigenphase labelling across the old degenerate sectors is used or asserted.

There is also a genuine extension of the earlier subcritical global transfer, but for **folded phases** rather than individually isolated clusters. Let epsilon_K be nonzero and tau_K=epsilon_K sqrt(2K) tend to zero. If exp(i theta_l) are the eigenvalues of U_(epsilon,K), define

$$y_l = \frac{\operatorname{Arg}(e^{iK\theta_l})}{\tau_K}, \qquad \operatorname{Arg} \in [-\pi, \pi).$$

Corollary B (subcritical folded Gaussian law). The empirical measures of y_l converge in quadratic Wasserstein distance to the standard normal law whenever epsilon_K sqrt(K) tends to zero.

Proof with a quantitative tail control. For two unitary matrices, the squared absolute overlaps of their orthonormal eigenbases form a doubly stochastic matrix. The corresponding average squared chord distance equals their normalized squared Hilbert–Schmidt difference. Decomposing this matrix into permutation matrices gives an eigenvalue matching with no larger average cost. The decomposition can be obtained by Hall's matching criterion and successive subtraction; it uses no eigenvalue simplicity. Thus (16), telescoped K times, matches the powered spectrum to exp(i tau_K x_l), where x_l are the eigenvalues of X_K, with root mean square chord error at most 6 tau_K^2/sqrt(K).

If |tau_K x_l|<=pi/2, principal argument differences obey |Arg u-tau_K x_l|<=4|u-exp(i tau_K x_l)|. On the complementary set |Arg u/tau_K-x_l|<=3|x_l|. Consequently

$$\begin{aligned} W_2(\operatorname{law}(y), \operatorname{law}(X_K)) &\le \frac{24|\tau_K|}{\sqrt{K}} + 3\sqrt{\tau_{-}\left(X_K^2 \mathbf{1}_{|X_K|>\pi/(2|\tau_K|)}\right)} \\ &\le \frac{24|\tau_K|}{\sqrt{K}} + \frac{6|\tau_K|}{\pi}\sqrt{\tau_{-}(X_K^4)}. \end{aligned} \tag{22}$$

The fourth moment stays bounded by Lemma 2, so the right side tends to zero. Combine with the W_2 conclusion of that lemma. End of proof.

The former isolated-cluster argument required epsilon_K K^(3/2) tending to zero. Corollary B permits an additional factor K in the kick scale, but its observable is a folded spectral measure. It must not be restated as an individual-cluster rank or no-overlap theorem.

7. Verification, reproducibility, and limits

The Python verifier constructs the cohomological generator explicitly, checks (4), compares the exact Fourier-support expression (12) to direct matrix multiplication, and compares all small odd-sector block spectra against the spectrum of X_K. Dense propagators and a separate chirp–FFT implementation agree numerically. The C++20 verifier works instead in the original integral T-coordinates with multiprecision integers; it verifies the signed-pair cardinalities and Pell arithmetic independently of the Python diagonal-coordinate representation.

The Lean module has general cleared-denominator identities for c_j and a finite original-coordinate K=11 certificate. Its compilation status must be checked separately. It does not formalize Duhamel estimates, Lemma 2, Wasserstein convergence, or Theorem A.

Extended numerical experiments estimate normalized traces with independent uniform fourth-root-of-unity vectors in an orthonormal odd basis. If R probes are used, the estimate is unbiased and its complex mean-square error is at most 1/(R d_N): the off-diagonal matrix-square sum is at most d_N for a unitary. The recorded square-root bound is an RMS bound over random probes, not a deterministic interval and not a 95 percent confidence interval. Fixed seeds ensure reproducibility. Both m=1 and m=2, including visible finite-size deviations, are retained. Finite agreement is not a proof of (A).

Reproduce the independent checks with CMake and the script `pell_floquet_regression.py` in the verification directory, passing `--cpp` and the compiled verifier path. The extended option is `--extended --probes 32`. Promoted evidence records source hashes and the scope of each check. The figure generator consumes that evidence rather than rerunning or editing selected points.

The ordinary two-factor product estimate is established mathematics. Tracial central limit theorems and semiclassically small perturbations of cat maps also have an existing literature. The arithmetic weighted-phase construction and the exact statement here require separate priority review; none is inferred from a few search results. See the companion source audit in `literature/pell_floquet_sources.md`.

Finally K is of order log N, whereas the odd-space Heisenberg scale is of order d_N. Equation (A) is a one-point spectral law of U_epsilon^K and a trace law at recurrence times. It supplies neither local correlations of U_epsilon nor fixed-strength classical BGS universality. The unresolved route requires joint trace control at times of order d_N, a justified symmetry reduction, and control under a nonvanishing classical perturbation.

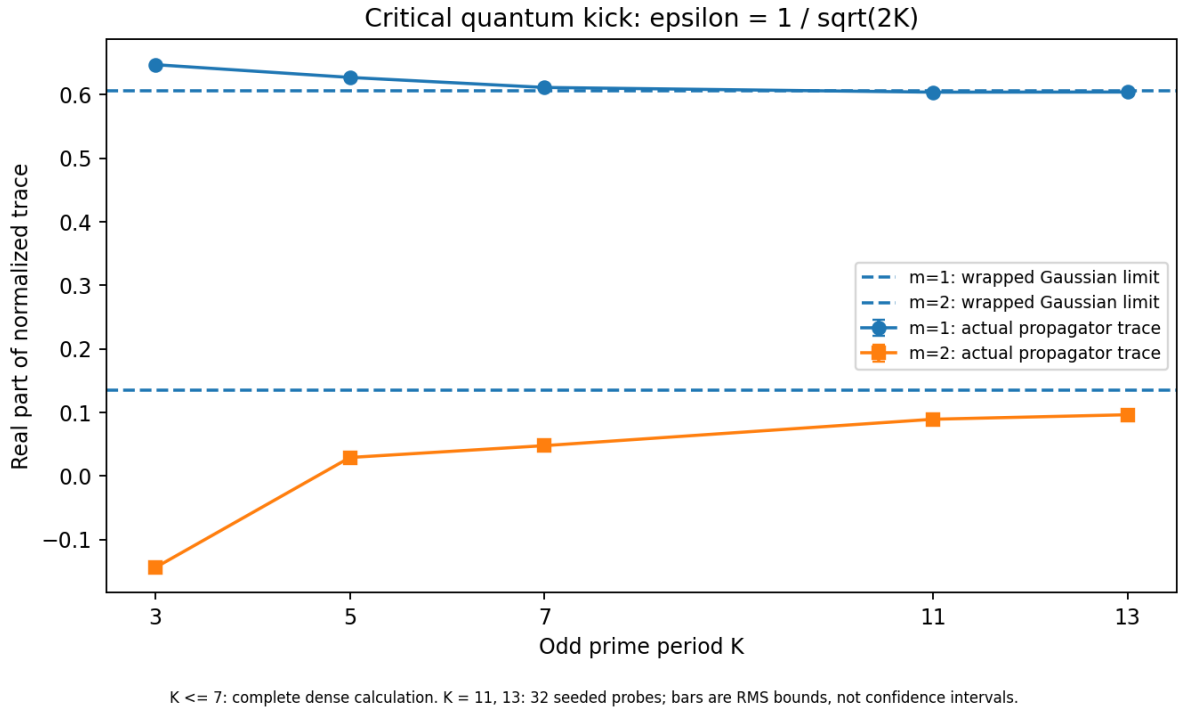
Related primary records

Greg Kuperberg, *A tracial quantum central limit theorem*, Transactions of the AMS 357 (2005), 459–471; arXiv:math-ph/0202035. The stated i.i.d. tracial setting is relevant precedent, not an assumption verified for our deterministic array.

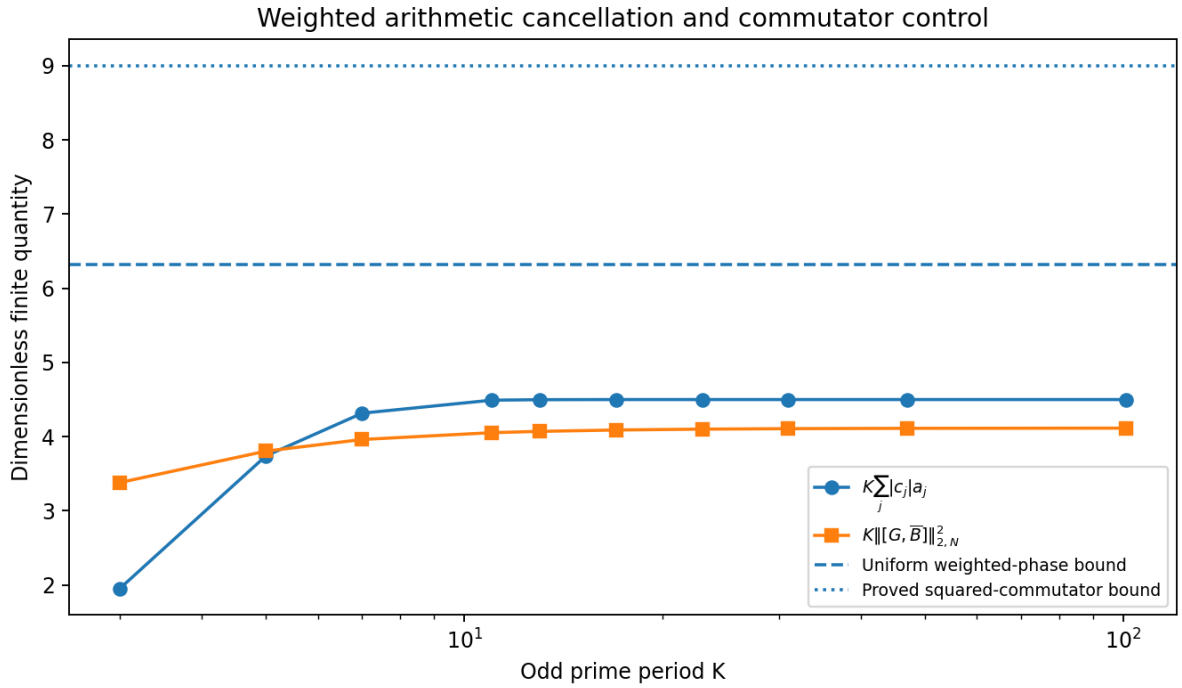
Theory of Trotter Error with Commutator Scaling, Physical Review X 11 (2021), 011020; DOI:10.1103/PhysRevX.11.011020. The two-factor estimate used above is standard; the arithmetic commutator bounds, rather than a new general product formula, are the specific input.

Martin Horvat and Mirko Degli Esposti, *Egorov property in perturbed cat map*, Journal of Physics A 40 (2007); arXiv:0706.4397. Its abstract concerns quantum–classical fidelity measured by phase-space overlap, a different statistic. This comparison uses primary abstracts and records, not a complete full-text priority review.

Reproducible numerical diagnostics



These are actual propagator traces, not fitted observable densities. Both Fourier modes are retained. Bars for the sampled traces represent an RMS bound over the random probes, not deterministic error bounds or confidence intervals. The visible finite-size bias is not removed.



The constant bounds are analytic; the points are independent finite evaluations. Exact pair-key checks use both diagonal coordinates (Python) and original integral map coordinates (C++20).